

SM-HYPONIC®

Hypoid Right Angle Gearmotor

Operating and Maintenance Manual

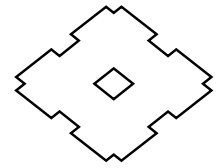


Manual

07.020.60.001

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MOUNTING

Solid Shaft Type

1. Mounting conditions

Ambient temperature: 14°F ~ 104°F

Humidity: 85% or less

Altitude: Lower than 3300 ft (1000 m) above sea level

Atmosphere: Free from corrosive gases, explosive gases or steam. It should also be free from dust and well ventilated.

Location: Indoors

2. Mount the gearmotor on a rigid surface.

3. There is no restriction for mounting angle.

4. Use hexagon socket head bolts when mounting RNFM series (flange-mount type). See **Table 1** for bolt sizes.

Table 1

Series	Frame Size	Size of hexagon socket head bolt
RNFM	20", 23"	M8
	30", 33"	M10
	40", 43"	M10
	50", 53", 54"	M12

Hollow Shaft Type

1. Mounting conditions

Ambient temperature: 14°F ~ 104°F

Humidity: 85% or less

Altitude: Less than 3300 ft (1000 m) above sea level

Atmosphere: Free from corrosive gases, explosive gases, steam and dust. It should also be well ventilated.

Location: Indoors

2. Mount the gearmotor on a driven shaft that has sufficient rigidity.

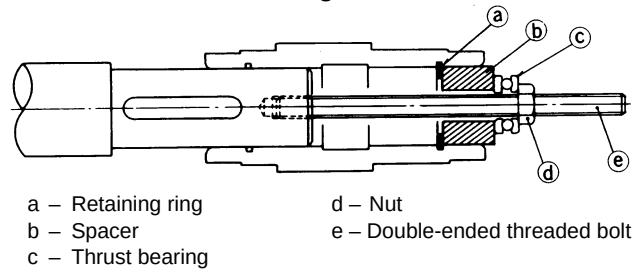
3. There is no restriction for the mounting angle.

4. Mounting Procedures

a. Connecting a Driven Shaft

Apply molybdenum disulfide grease to the surface of the shaft and the inner surface of the hollow shaft. Slide the SM-Hyponic onto the shaft. To make the installation smoother if the fit is too tight, lightly tap the end of the hollow output shaft with a wooden hammer. **Avoid hitting the casing.** To ensure smooth installation of the drive, we recommend the use of a jig shown in Fig. 1.

Fig. 1



The hollow shaft is made according to ISO H8 tolerances. Following installation, ensure that the fitting between the hollow and the driven shaft is tightened correctly. We recommend ISO js6 or k6 as the tolerance for the driven shaft.

b. Mounting the SM-Hyponic gearmotor

Fig. 2 Stepped Shaft Option

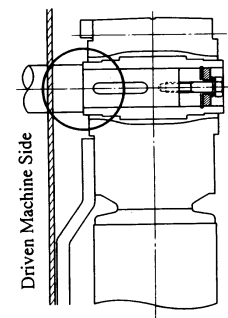


Fig. 3 Spacer Option

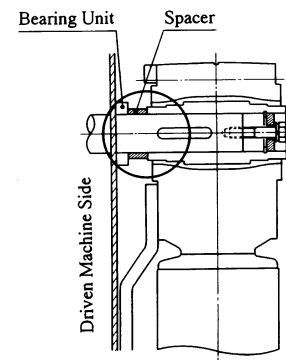
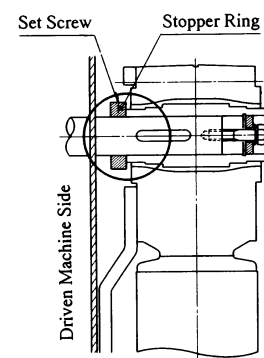


Fig. 4 Set Screw Option



MOUNTING

Fig. 5 Spacer and Plate Option

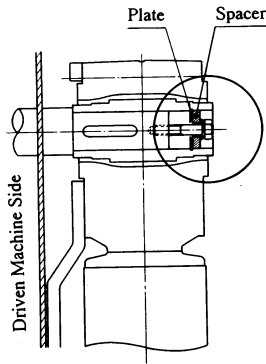


Fig. 6 End Plate Option

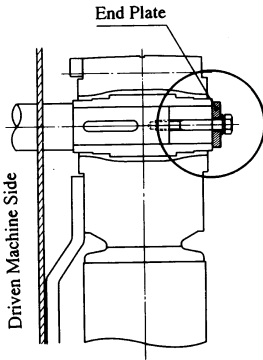
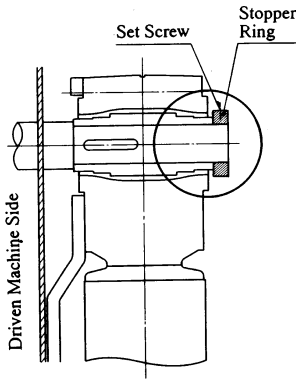


Fig. 7 Set Screw and Stopper Ring Option



The torque arm (Section A in Fig. 8) should be mounted to ensure that the contact surface between the drive and shaft are free from excessive forces. **Do not attach the torque arm using anti-rotation bolts.**

For applications that require frequent starts and stops or frequent reversing, insert a rubber bushing between the torque arm and securing bolt (or spacer) in order to dampen impact load.

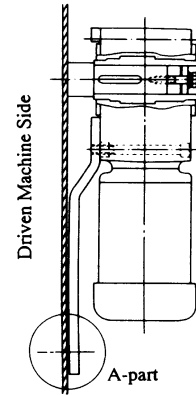
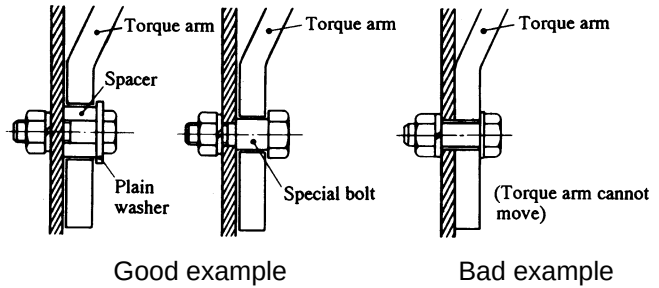


Fig. 8 Torque Arm Securing Methods

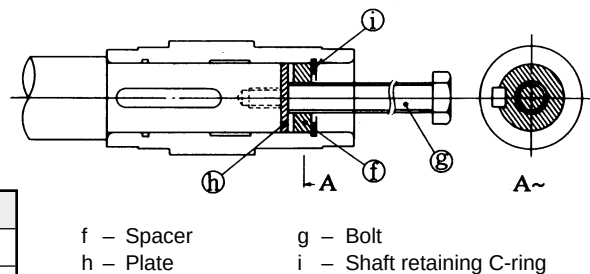


(d) Removing the shaft.

Do not apply excessive force to the gearmotor and shaft. Using a jig as shown in Fig. 9 will facilitate removal of the shaft.

Note: The customer should prepare parts for setting, securing or removing the shaft.

Fig. 9

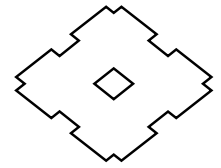


(c) Installing the torque arm.

Mount the torque arm on the driven machine side of the drive casing. Use hexagon socket head bolts for mounting. (See Table 2 for bolt sizes.)

Table 2

Series	Frame Size	Size of hexagon socket head bolt
RNFM	20, 23	M8
	30, 33	M10
	40", 43"	M10
	50", 53", 54"	M12
	60, 63, 64	M20

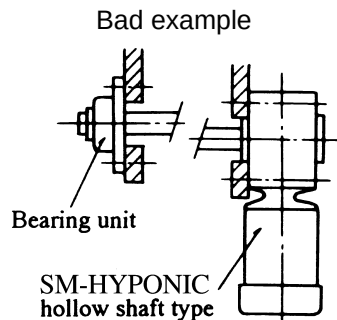
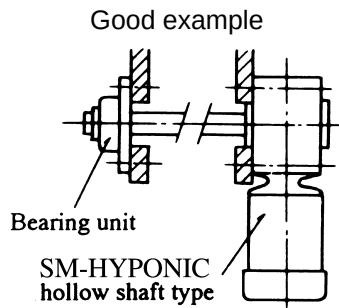


MOUNTING

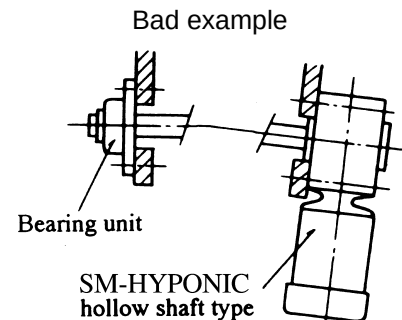
(5) Flange and Foot Mounting (optional).

When installing the SM-Hyponic, ensure that the gearmotor and the shaft of the driven machine are properly aligned so that the drive is free from excessive force.

Fig. 10 Flange coupling

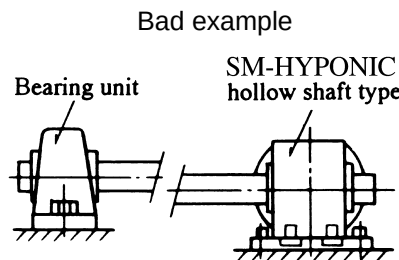
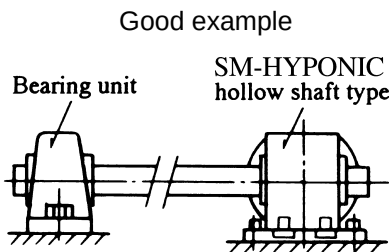


(The concentricity between the shaft and mounting pilot is out of allowable range.)

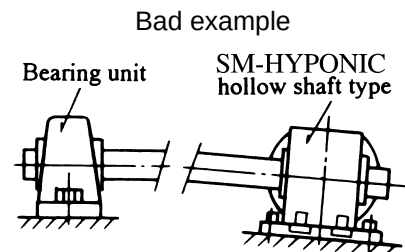


(The shaft centerline is not positioned at right angles to the flange.)

Fig. 11 Foot mounting (optional)



(The shaft center of the bearing unit does not align with that of the Drive.)



(The parallelism of the mounting beds is out of allowable range.)

CONNECTING TO THE DRIVEN MACHINE

Solid-shaft type

1. Mount the connecting device, such as a coupling, chain, sprocket, gear or V-pulley, on the shaft as close as possible to the shaft collar. This places the load point between the center of the shaft and the shaft collar.
2. We recommend using end cap screws to avoid possible bearing damage from excessive force or thrust load that may be applied to the shaft while fitting the connecting device.
3. When connecting the SM-Hyponic gearmotor to the driven machine, be sure to align the shafts of both units (for coupling connection) or keep both shafts parallel (for chain, gear or V-belt connection).
4. Excessively loose chains will cause a jolt upon start-up that may damage the SM-Hyponic gearmotor and the driven machine. Excessive tension of V-belts may cause bearing failure.

Fig. 12

Gear, chain, sprocket

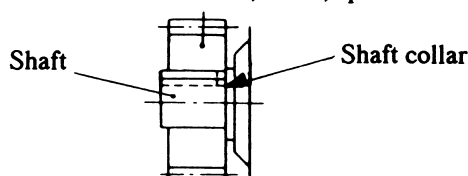
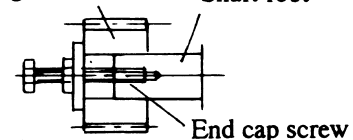


Fig. 13

Connecting device

Shaft foot



WIRING

1. Prior to wiring, refer to the name plate mounted to the motor portion of the SM-Hyponic gearmotor. Check the power supply, interconnects, relays, protective starting devices (i.e., Star (Wye) delta – if reduced voltage starting is required), space heaters, thermal sensors and other accessories.
2. Be aware that long wiring may cause voltage drops.
3. Figures 14a – d show standard specifications for wire connections and terminal marks. Figures 15a – d show specifications for inverter connections. Figure 16 shows the rotating direction of the output shaft when using wiring connections shown in Figures 14a – d. Table 3 lists Brake lining size specifications and Table 4 lists Varistor Specifications.

Standard Wiring Connection, Dual Voltage

Fig. 14-a Normal Brake Action, High Voltage

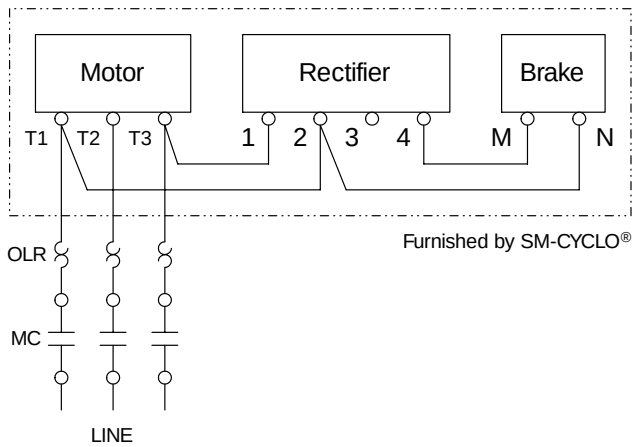


Fig. 14-b Fast Brake Action, High Voltage

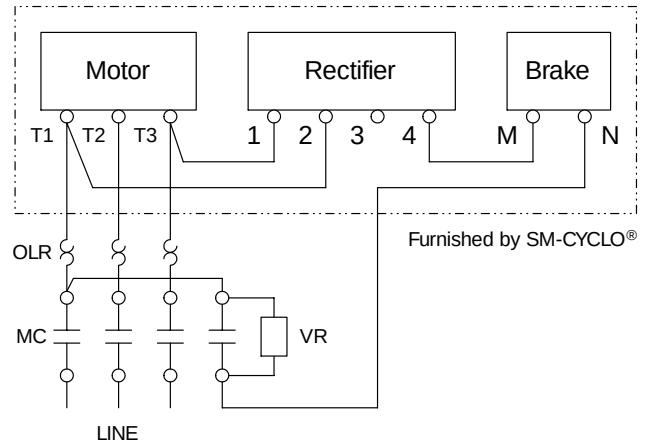


Fig. 14-c Normal Brake Action, Low Voltage

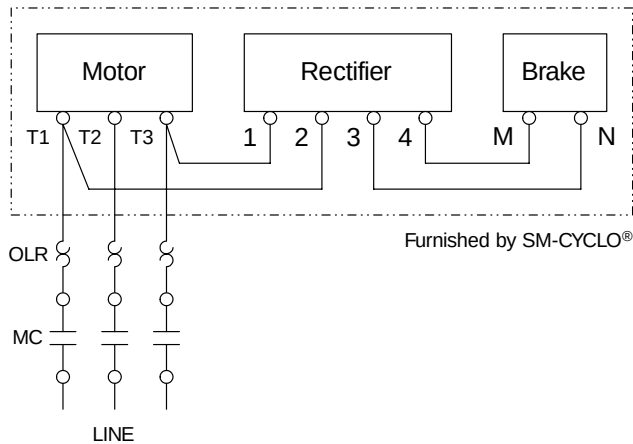
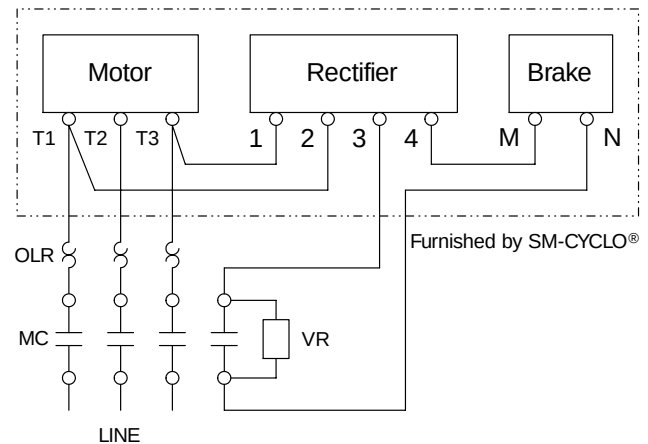
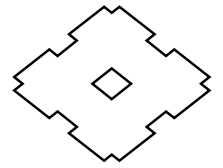


Fig. 14-d Fast Brake Action, Low Voltage





Inverter Wiring Connection, Dual Voltage

Symbols

MC: Electromagnetic contactor

MCB: Magnetic circuit breaker

OLR: Overload or thermal relay

VR: Varistor (protective device)

Fig. 15-a Normal Brake Action, High Voltage

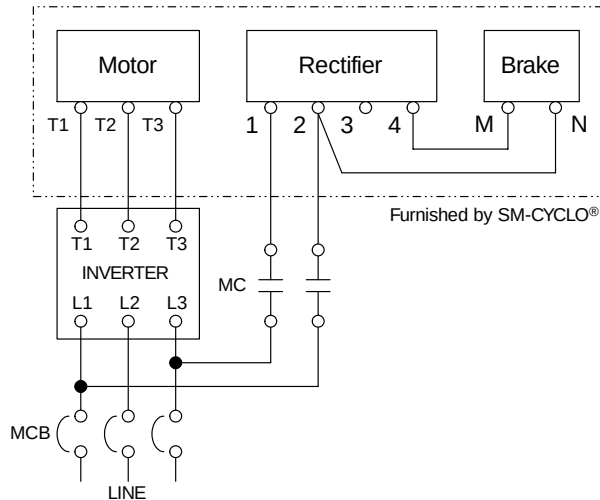


Fig. 15-b Fast Brake Action, High Voltage

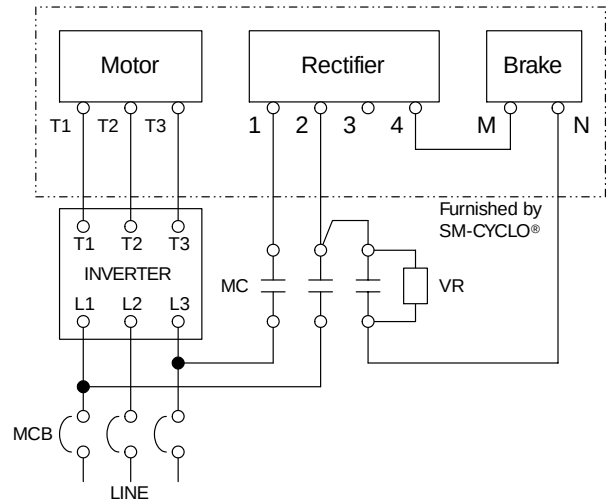


Fig. 15-c Normal Brake Action, Low Voltage

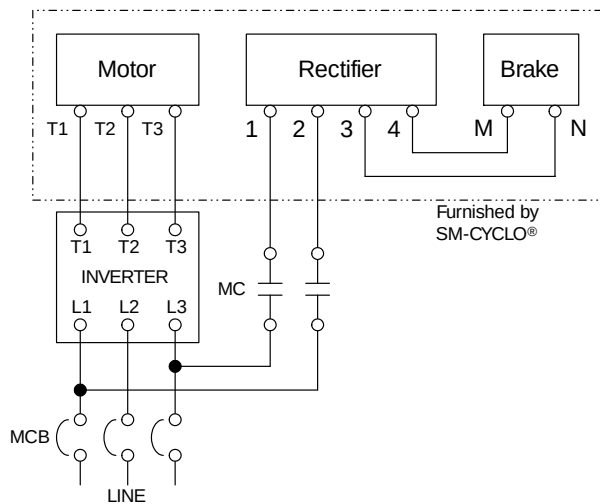
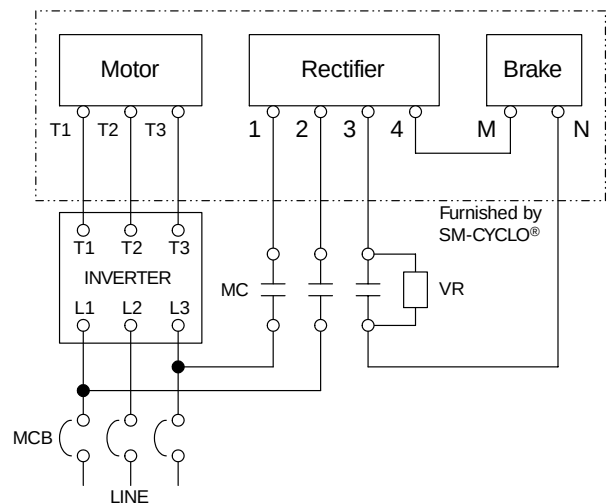


Fig. 15-d Fast Brake Action, Low Voltage



Note: Recommended brake contactor size for fast acting circuit is greater than 5 times the rated current shown in Table 8 on page 8.

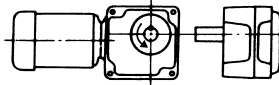
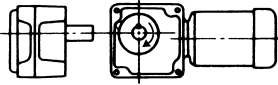
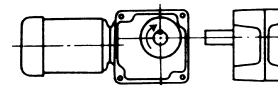
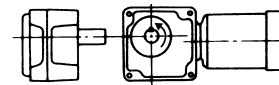
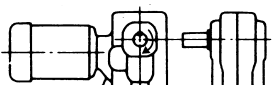
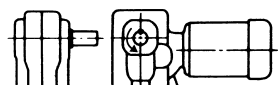
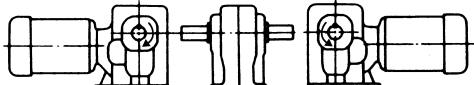
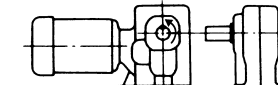
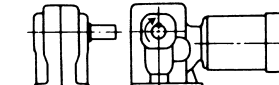
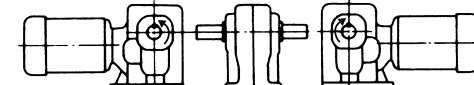
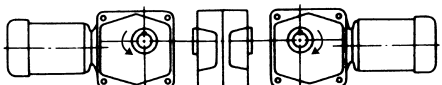
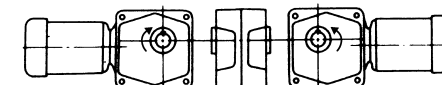
Table 4 Varistor Specifications

Operating Voltage		200-230V	380-460V
Var. Rated Voltage		AC260~300V	AC510V
Varistor Voltage		430~470V	820V
Rated Watt	FB-01A, 02A	Over 0.2W	Over 0.4W
	FB-05A	Over 0.2W	Over 0.4W

WIRING

Fig. 16 Rotating Direction of the Output Shaft

If wiring connection has been completed as shown in Fig. 14, a motor shaft rotates clockwise as seen from the fan cover side. The rotating directions of an output shaft are indicated by arrows in Table below.							
kW	HP	Frame Size	Speed ratio	kW	HP	Frame Size	Speed reduction ratio
0.1	1/4	20	10 • 12 • 15 • 20 • 25 • 30 • 40 • 50 • 60	0.1	1/4	20	80 • 100 • 120
0.2	1/3	23	10 • 12 • 15 • 20 • 25 • 30	0.2	1/3	23	40 • 50 • 60
		30	10 • 15 • 20 • 30 • 40 • 50 • 60			30	80 • 100 • 120
0.4	1/2	33	10 • 12 • 15 • 20 • 25 • 30	0.4	1/2	33	40 • 50 • 60
		40	10 • 15 • 20 • 30 • 40 • 50 • 60			40	80 • 100 • 120
0.75	3/4	43	10 • 12 • 15 • 20 • 25 • 30	0.75	3/4	43	40 • 50 • 60
		50	10 • 15 • 20 • 30 • 40 • 50 • 60			50	80 • 100 • 120
1.5	1.5	53	10 • 12 • 15 • 20 • 25 • 30	1.5	1.5	53	40 • 50 • 60 • 80
		60				60	80 • 100 • 120
2.2	3	54	10 • 12 • 15 • 20 • 25 • 30	2.2	3	54	40 • 50 • 60
		60	10 • 12 • 15 • 20 • 25 • 30 • 40 • 50			60	60 • 80
3.7	5	63	10 • 12 • 15 • 20 • 25 • 30	3.7	5	63	40 • 50
5.5	7½	64	10 • 12 • 15 • 20 • 25	5.5	7½	64	30

RNFM series  R type  L type	RNFM series  R type  L type
RNHM series  R type  L type  T type	RNHM series  R type  L type  T type
RNYM series 	RNYM series 

Note: Replacing two of the three power supplies of a three-phase induction motor will cause the motor to rotate in directions reverse to those shown in above Figure. Also replacing X with Y of single-phase induction motor will cause the motor to rotate in directions reverse to those shown in above Figure.

4. Motor Operation Precautions

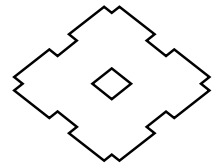
- Always ground the motor terminal box or frame.**
- Insulation resistance

Determine the insulation resistance (to do this, separate the motor and the control board). The insulation resistance value will vary depending on the temperature, humidity, extent of contamination, the servicing period, test running

time, as well as the motor output, voltage and type of insulation. Therefore, the insulation resistance (r) cannot be expressed uniformly; however, it should be equal to or greater than the value listed in Table 5.

Table 5 Insulation Resistance

Motor voltage	Mega voltage	Insulation resistance (R)
Low voltage motor (Lower than 600V)	500V	Higher than 1 M Ω



LUBRICATION

1. SM-Hyponic gearmotors are grease-lubricated. They are filled prior to shipment and arrive ready for customer use.
2. Ensure that the connection to the driven machine is correct.
3. Ensure that the rotation direction is correct.
4. After completing these steps, start the test run, without any load, and increase the load gradually. Also observe the precautions listed in Table 6.

OPERATION

Once the SM-Hyponic gearmotor is installed, ensure that the wiring is correct and secure prior to operation. Observe the precautions listed in Table 6 during the trial run. Stop running the gearmotor if any abnormalities are detected and contact your nearest sales office or distributor.

Table 6 Trial Run Precautions

Observation	Possible Cause
Any abnormal noises or vibrations.	(1) The casing is distorted due to an uneven mounting surface. (2) The gearmotor is resonant due to insufficient rigidity of the mounting surface. (3) The shaft of the SM-Hyponic and the driven machine are not aligned. (4) Vibrations of the driven machine are conveyed to the gearmotor. (5) Rigidity of the driven machine and its shaft is insufficient (hollow shaft type). (6) Excessive force is exerted on the baffle sections of the torque arm (hollow shaft type). (7) After the SM-Hyponic gearmotor is flange or foot mounted, undue force is exerted between the gearmotor and its shaft.
Abnormally high temperature of the gear casing or motor frame surface.	(1) The electric current is exceeding the rated value specified on the nameplate. (2) The rise and fall of the electric current is too intense. (3) The ambient temperature of the gearmotor is too high.

DAILY INSPECTION AND MAINTENANCE

1. Inspect the gearmotor daily for each of the items listed in Table 7.
2. If any abnormalities are found during daily inspections, follow the procedures outlined in the Troubleshooting Guide, Table 20 on page 14. If the abnormality is not listed or the recommended procedure does not solve the problem, contact your nearest sales office or distributor.
3. The SM-Hyponic gearmotor does not require grease replenishment, but overhauling after 10,000 hours of operation, or 2 – 3 years will extend its life. Note: Over-hauling consists of disassembling the unit, replacing the seals and gaskets, cleaning the internal parts and then repacking the unit with designated grease.

Table 7 Daily Inspection Items

Items	Details
Electric current	Is the electric current higher than the value specified on the nameplate?
Noise	Is the gearmotor making any unusual noises?
Vibration	Are there any unusual vibrations of the gear case or motor frame?
Surface temperature	Is the surface temperature of the gear case, or motor frame not too high or rising suddenly? (The temperature rise during operation varies depending on the type of motor. There may be a problem, if the temperature frequently rises approximately 40°C (104°F) above the ambient temperature.)
Grease leakage	Is there any grease leaking from the gear assembly?
Mounting bolts	Are there any loose mounting bolts?
Chains and V-belts	Are there any loose chains or V-belts?
Brake	Is the brake lining worn out?

FB BRAKE ASSEMBLY – INSPECTION, ADJUSTMENT & MAINTENANCE

SM-Cyclo FB series brakemotors are designed to be mechanically rugged, electrically reliable and efficient in operation. To maintain this reliable performance, the brake assembly must be inspected and adjusted periodically.

This section of the manual pertains specifically to the brake portion of the SM-Hyponic gearmotor and provides all the necessary information to insure long and trouble-free service.

Brake Models FB-01A, FB-02A and FB-05A

1. Standard Brakemotor Specifications

Table 8 lists the standard specifications for Models FB-01A, FB-02A and FB-05A.

Table 8 Models FB-01A, FB-02A, FB-05A Standard Specifications

Brake Type	Motor HP	Brake Torque ft-lb	Inertia WK ² lb-ft ²	Brake Coil	Brake Current (A)		Coil Resis ohms	Brake Delay Time (seconds)	
					230V	460V		Normal	Fast
FB-01A	1/8	0.7	0.0083	DC Energized Type, Built-in Rectifier within Conduit Box	0.1	0.06	2700	0.15 ~ 0.2	0.015 ~ 0.02
FB-02A	1/4 1/3	1.4	0.0131		0.1	0.06	1791	0.15 ~ 0.2	0.015 ~ 0.02
FB-05A	1/2	2.9	0.016		0.1	0.06	1791	0.1 ~ 0.15	0.01 ~ 0.015

Notes: 1) Continuous time rating for both the brake and motor.
2) Indoor types can be installed for use in any orientation.

2. Construction and Operating Principles

a. Construction

Fig. 17 illustrates the construction of the brake. The restraining bolt (4) fastens the brake shoe (10) and spacer (2) onto the stationary core (1). The armature plate (11) is kept from rotation by the restraining bolt (4) but moves axially by electromagnetic attraction and the tension of the pressure spring (12). The brake lining (3) is fitted to the hub (5), which is secured to the motor shaft with a key. The solenoid coil (13) is energized via a rectifier located within the conduit box.

b. Operating Principles

The brake is a (fail-safe type) spring actuated type brake that releases the brake mechanism when the solenoid coil is energized and engages when the solenoid coil is not energized.

When power is applied to the unit, the solenoid coil and electric motor become energized and the energized coil attracts the armature plate (11) against the tension of the pressure spring (12). As a result, the brake lining (3) disengages and the motor starts to run.

When the power is disconnected, the solenoid coil and electric motor are not energized. This causes the pressure spring (12) to actuate the armature plate (11), which in turn presses the brake lining (3) against the brakeshoe (10) and brings the motor to a quick stop.

Fig. 17 FB-01A, FB-02A, FB-05A Models

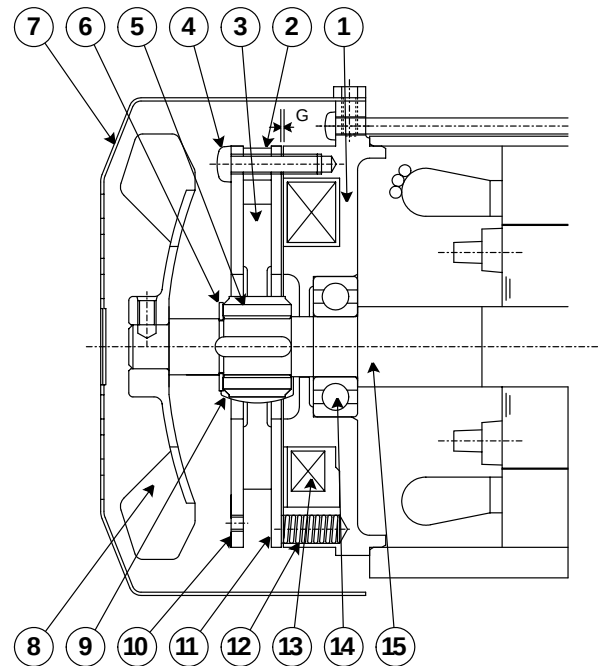
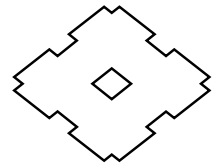


Table 9 FB-01A, -02A, -05A Parts

No.	Part Name	No.	Part Name
1	Stationary Core*	9	Leaf Spring*
2	Spacer*	10	Brake Shoe*
3	Brake Lining*	11	Armature*
4	Restraining Bolt*	12	Pressure Spring*
5	Hub*	13	Solenoid Coil*
6	C-type Retaining Ring	14	Ball Bearing
7	Cover	15	Motor Shaft
8	Fan (TEFC model only)		

*These parts are included in a complete brake kit.



3. Inspection

At regular intervals, check that:

- the unit is operating normally.
- the brake lining is not excessively worn (or gap G is normal).
- all the mounting screws are securely tightened.

4. Gap Inspection

The brake lining will wear after the unit has been used for a long period of time. Regularly check that gap G (Fig. 17) is at an acceptable value. If gap G becomes too large, the solenoid coil may fail to pull in the armature plate, and hence cannot release the brake, resulting in the unit remaining in a continuously braked condition. Follow these steps to inspect the brake gap:

- Remove the cover (7).
- Insert a gap gage into the space between the stationary core (1) and armature plate (11). Measure the gap size at three appropriate circumferential points.
- The gap needs to be adjusted if the values are close to the allowable limit listed in Table 10.

Table 10 Brake Gap Size

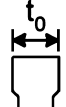
Brake Type	Gap value G (in)	
	Spec. value	Allowable limit
FB-01A	0.006 ~ 0.010	0.020
FB-02A		
FB-05A		

5. Gap Adjustment

If the brake lining is so heavily worn that gap adjustment is required, follow these steps:

- Remove the cover (7).
- Loosen the restraining bolt (4), rotate the brake shoe one complete turn counterclockwise and retighten the restraining bolt (4). After tightening the restraining bolt, measure the gap G to verify that it falls between the specification value and the allowable limit shown in Table 10. (This procedure reduces the gap approximately 0.012 inch.)
- Turn the system power on and off a few times to check the brake performance.
- Replace the cover (7).

Table 11 Brake Lining Size

Brake Type	Brake lining dimension	Initial thickness t_0 (in)
FB-01A		0.276
FB-02A		
FB-05A		

Brake Models FB-1B, FB-2B and FB-3B

1. Standard Brakemotor Specifications

Table 12 lists the standard specifications for Models FB-1B, FB-2B and FB-3B.

Table 12 Models FB-1B, FB-2B, FB-3B Standard Specifications

Brake Type	Motor HP	Brake Torque ft-lb	Inertia WK ² lb-ft ²	Brake Coil	Brake Current (A)		Coil Resis ohms	Brake Delay Time (seconds)	
					230V	460V		Normal	Fast
FB-1B	3/4 1	5.8 5.8	0.0267 0.0308	DC Energized Type, Built-in Rectifier within Conduit Box	0.1	0.06	1470	0.2 ~ 0.3	0.01 ~ 0.02
FB-2B	1.5 2	11 11	0.0504 0.0558		0.3	0.2	589	0.2 ~ 0.3	0.01 ~ 0.02
FB-3B	3	16	0.0884		0.3	0.2	589	0.3 ~ 0.40	0.01 ~ 0.02

Notes: 1) Continuous time rating for both the brake and motor.
2) Indoor types can be installed for use in any orientation.

FB BRAKE ASSEMBLY – INSPECTION, ADJUSTMENT & MAINTENANCE

2. Construction and Operating Principles

a. Construction

Fig. 17 illustrates the construction of the brake. The restraining bolt (7) fastens the brake shoe (15), gap adjusting shim (5) and spacer (4) onto the stationary core (1). The restraining bolt (7) keeps the armature plate from rotating, but the plate moves axially by electromagnetic attraction and the tension of the pressure spring (17). The brake lining (8) is fitted to the hub (10), which is secured to the motor shaft with a key. The solenoid coil (18) is energized via a rectifier located within the terminal box.

Fig. 18 FB-1B, -2B, -3B Models

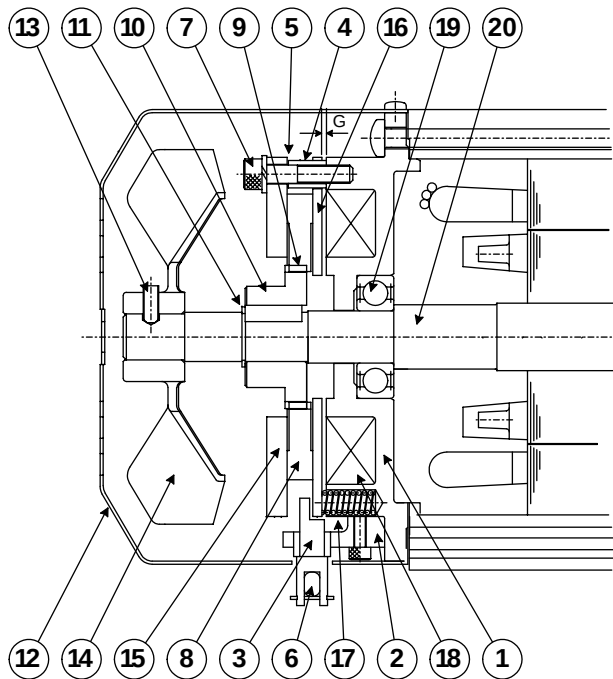


Table 13 FB-1B, -2B, -3B Parts

No.	Part Name	No.	Part Name
1	Stationary Core*	11	Retaining Ring
2	Brake Release Support	12	Fan Cover
3	Shifting Pin	13	Fan Set Pin
4	Spacer*	14	Fan
5	GAP Adjusting Sleeve*	15	Brake Shoe*
6	Brake Release Lever	16	Armature*
7	Restraining Bolt*	17	Pressure Spring*
8	Brake Lining*	18	Solenoid Coil*
9	Leaf Spring*	19	Fan Side Bearing
10	Hub*	20	Motor Shaft

*These parts are included in a complete brake kit.

b. Operating Principles

The brake is a (fail-safe type) spring actuated type brake that releases the brake mechanism when the solenoid coil is energized and engages when the solenoid coil is not energized.

When power is applied to the unit, the solenoid coil and electric motor become energized and the energized coil attracts the armature plate (16) against the tension of the pressure spring (17). As a result, the brake lining (8) disengages and the motor starts to run.

When the power is disconnected, the solenoid coil and electric motor are not energized. This causes the pressure spring (17) to actuate the armature plate (16), which in turn presses the brake lining (8) against the brakeshoe (15) and brings the motor to a quick stop.

3. Inspection

a. At regular intervals, check that:

- the unit is operating normally.
- the brake lining is not excessively worn (or gap G is normal).
- all the mounting screws are securely tightened.

b. Manual brake release procedure

FB-1B, -2B, -3B brakemotors are equipped with a one-touch release mechanism. To manually release the brake with power to the unit turned off, pull the brake release lever out from its holder and push it forward toward the reducer. Releasing the lever will re-engage the brake.

4. Gap Inspection

The brake lining will wear after the unit has been used for a long period of time. Regularly check that gap G (Fig. 18) is at an acceptable value. If gap G becomes too large, the solenoid coil may fail to pull in the armature plate, and hence cannot release the brake, resulting in the unit remaining in a continuously braked condition. Follow these steps to inspect the brake gap:

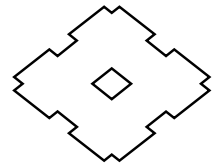
a. Remove the cover (12).

b. Insert a gap gage into the space between the stationary core (1) and armature plate (16). Measure the gap size at three appropriate circumferential points.

c. The gap needs to be adjusted if the values are close to the allowable limit listed in Table 14.

Table 14 Brake Gap Size

Brake Type	Gap value G (in)	
	Spec. value	Allowable limit
FB-1B	0.008 ~ 0.012	0.020
FB-2B	0.008 ~ 0.012	0.020
FB-3B	0.008 ~ 0.012	0.028



5. Gap Adjustment

If the brake lining is so heavily worn that gap adjustment is required, follow these steps:

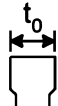
- Remove the cover (12). Measure the gap size to confirm the deviation from the specification value. The minimum adjustable setting is no less than the thickness of the Gap adjusting shim, 0.008 in.
- Loosen the set pin (13) and remove the fan (14).
- Slightly loosen the restraining bolt (7) and remove parts (4), (5), (7) and (15) as a set. Be careful not to remove only the bolt (7) and lose the shims (5).
- One gap adjusting shim (5) is 0.008 in. thick. Decrease the number of shims in use according to the degree of wear (**Note: Retain the removed shims for use during the brake lining replacement procedure**). Reassemble parts (4), (5), (7) and (15) as a set.
- Once reassembled, check gap G. If the gap size is still too large, adjust the number of shims again.
- After completing the gap adjustment, turn the system power on and off a few times to check the brake performance.
- Replace the fan (14), set pin (13) and cover (12).

6. Brake Lining Replacement

Follow these steps to replace the brake lining when its thickness has reached the allowable limit shown in Table 15, or when sleeve adjustment is no longer an effective means of gap adjustment:

- Remove the cover (12) and measure the gap G. Remove the set pin (13) and the fan (14).
- Slightly loosen the restraining bolt (7) and remove parts (4), (5), (7) and (15) as a set.
- Remove the brake lining (8), taking care to prevent the leaf spring from coming off.
- Install the new brake lining, taking care not to damage or remove the leaf spring (9). Ensure that the lining moves smoothly along the hub (10).
- Replace any gap adjusting shims removed and retained from previous gap adjustments. Then reinstall parts (4), (5), (7) and 15 as a set.
- Measure gap G. Readjust if the gap is not within the specification value range.
- Turn the system power on and off a few times to check the brake performance. If no abnormalities are detected, replace the fan (14), set pin (13) and cover (12).

Table 15 Brake Lining Size

Brake Type	Brake lining dimension	Initial thickness t_0 (in)	Allowable thickness limit t_0 (in)
FB-1B		0.276	0.236
FB-2B		0.322	0.283
FB-3B		0.354	0.315

Brake Models FB-5B and FB-8B

1. Standard Brakemotor Specifications

Table 16 lists the standard specifications for Models FB-5B and FB-8B.

Table 16 Models FB-5B, FB-8B Standard Specifications

Brake Type	Motor HP	Brake Torque ft-lb	Inertia WK^2 lb-ft ²	Brake Coil	Brake Current (A)		Coil Resis ohms	Brake Delay Time (seconds)	
					230V	460V		Normal	Fast
FB-5B	5	27	0.0227	DC Energized Type, Built-in Rectifier within Conduit Box	0.7	0.3	308	0.4 ~ 0.5	0.01 ~ 0.02
FB-8B	7.5	40	0.0297		0.7	0.3	308	0.3 ~ 0.4	0.01 ~ 0.02

Notes: 1) Continuous time rating for both the brake and motor.
2) Indoor types can be installed for use in any orientation.

FB BRAKE ASSEMBLY – INSPECTION, ADJUSTMENT & MAINTENANCE

2. Construction and Operating Principles

a. Construction

Fig. 19 illustrates the construction of the brake. Among the brake parts, the stationary core (1), solenoid coil (18), and stud bolt (3) constitute an integral subassembly unit. The stud bolt (3) keeps the armature plate (16) from rotating, but the plate moves axially by electromagnetic attraction and the tension of the pressure spring (17). The adjusting washer (4) and spring washer (7) hold the brake shoe (15) against the nut (8) at all times. The brake lining (9) is fit to the hub (10), which is secured to the motor shaft with a key.

Fig. 19 FB-5B, FB-8B Models

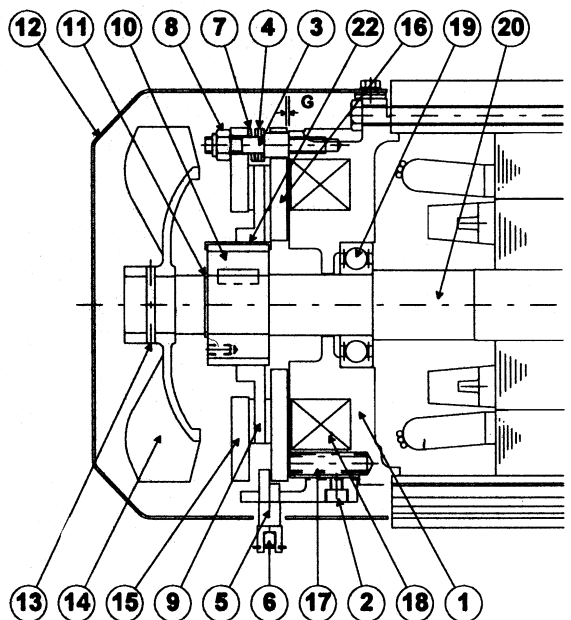


Table 17 FB-5B, -8B Parts

No.	Part Name	No.	Part Name
1	Stationary Core*	12	Fan Cover
2	Brake Release Support	13	Fan Set Screw or Pin
3	Stud Bolt*	14	Fan
4	GAP Adjusting Washer*	15	Brake Shoe*
5	Shifting Pin	16	Armature*
6	Brake Release Lever	17	Pressure Spring*
7	Spring Washer*	18	Solenoid Coil*
8	Nut*	19	Fan Side Bearing
9	Brake Lining*	20	Motor Shaft
10	Hub*	21	Bearing Cover
11	Retaining Ring	22	Leaf Spring*

*These parts are included in a complete brake kit.

b. Operating Principles

The brake is a (fail-safe type) spring actuated type brake that releases the brake mechanism when the solenoid coil is energized and engages when the solenoid coil is not energized.

When power is applied to the unit, the solenoid coil and electric motor become energized and the energized coil attracts the armature plate (16) against the tension of the pressure spring (17). As a result, the brake lining (9) disengages and the motor starts to run.

When the power is disconnected, the solenoid coil and electric motor are not energized. This causes the pressure spring (17) to actuate the armature plate (16), which in turn presses the brake lining (9) against the brakeshoe (15) and brings the motor to a quick stop.

3. Inspection

a. At regular intervals, check that:

- the unit is operating normally.
- the brake lining is not excessively worn (or gap G is normal).
- all the mounting screws are securely tightened.

b. Manual brake release procedure

FB-5B, -8B brakemotors are equipped with a one-touch release mechanism. To manually release the brake with power to the unit turned off, pull the brake release lever out from its holder and push it forward toward the reducer. Releasing the lever will re-engage the brake.

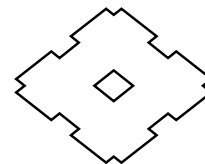
4. Gap Inspection

The brake lining will wear after the unit has been used for a long period of time. Regularly check that gap G (Fig. 18) is at an acceptable value. If gap G becomes too large, the solenoid coil may fail to pull in the armature plate, and hence cannot release the brake, resulting in the unit remaining in a continuously braked condition. Follow these steps to inspect the brake gap:

- Remove the cover (12).
- Insert a gap gage into the space between the stationary core (1) and armature plate (16). Measure the gap size at three appropriate circumferential points.
- The gap needs to be adjusted if the values are close to the allowable limit listed in Table 18.

Table 18 Brake Gap Size

Brake Type	Gap value G (in)	
	Spec. value	Allowable limit
FB-5B	0.016 ~ 0.020	0.039
FB-8B	0.016 ~ 0.020	0.039

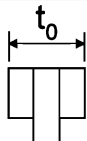


5. Gap Adjustment

If the brake lining is so heavily worn that gap adjustment is required, follow these steps:

- Remove the cover (12).
- Insert a gap gage into the space between the stationary core (1) and the armature plate (16) and rotate the nut (8) at the tip of the stud bolt (3) clockwise until the gap measures an appropriate size. If the gap is too large to adjust by this procedure, decrease the number of adjusting washers (4) in use. Evenly adjust the three nuts (8) until the gaps at the three circumferential points are equal and fall within the specification range shown in Table 18.
- After completing the gap adjustment, turn the system power on and off a few times to check the brake performance.
- Replace the fan (14), set pin or screw (13) and cover (12).

Table 19 Brake Lining Size

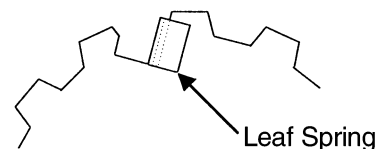
Brake Type	Brake lining dimension	Initial thickness t_0 (in)	Allowable thickness limit t_0 (in)
FB-5B		0.394	0.237
FB-8B		0.394	0.237

6. Brake Lining Replacement

Follow these steps to replace the brake lining when its thickness has reached the allowable limit shown in Table 19, or when sleeve adjustment is no longer an effective means of gap adjustment:

- Remove the cover (12), set pin (13) and fan (14).
- Remove all three nuts (8)
- Remove the brake shoe (15) and take out the brake lining (9).
- Fix the leaf spring (22) as shown in Fig. 20.
- Apply a small amount of grease along the spline of the new brake lining (9), taking care not to apply any to the wear surface.
- Fit the new brake lining (9) onto the hub (10) and check that it moves smoothly. Remove any excess grease.
- After reassembling the brake, measure gap G. If the gap is out of the specification range, adjust by rotating the gap adjusting nut (8).
- Turn the system power on and off a few times to check the brake performance. If no abnormalities are detected, replace the fan (14), set pin (13) and cover (12).

Fig. 20 Leaf Spring



TROUBLESHOOTING

The SM-Hyponic is running normally when it meets the following criteria:

- The motor begins to run immediately after the start switch is moved to the ON position.
- The unit does not make any abnormal sounds during operation.
- The motor stops running within about 0.5 seconds after power to the unit is switched off.

If you find any abnormality, refer to Table 20 Quick Troubleshooting Guide on pages 14 – 15, and take the appropriate corrective action as soon as possible.

TROUBLESHOOTING

Table 20 Quick Troubleshooting Guide

Problem			Possible Cause	Corrective Action
SM-HYPONIC	Runs Hot	Overloading	Load exceeds the capacity of the SM-Hyponic	Check rated capacity of the SM-Hyponic; replace with unit of sufficient capacity or reduce load.
		Improper lubrication	Insufficient lubrication	Check lubricant level and increase to recommended level.
			Excessive lubrication	Check lubricant level and reduce to recommended level.
			Wrong lubricant	Flush out and refill with correct lubricant as recommended.
	Vibration or noise	Loose Foundation bolts	Weak mounting structure	Inspect mounting of SM-Hyponic. Tighten loose bolts and/or reinforce mounting structure.
			Loose bolts	Tighten bolts.
		Failure of bearings	May be due to lack of lubricant	Replace bearing(s). Clean and flush SM-Hyponic; fill with recommended lubricant.
			Overload	Check rated capacity of SM-Hyponic, replace with unit of sufficient capacity or reduce load.
		Insufficient lubricant	Level of lubricant in the SM-Hyponic not properly maintained	Check lubricant level and adjust to factory recommended level.
	Output shaft does not turn	Motor shaft broken	Overloading of reducer can cause damage	Replace broken shaft. Check rated capacity of SM-Hyponic.
			Key missing or sheared off on input shaft	Replace key.
		Motor doesn't turn	Motor	Refer to Motor section of Troubleshooting Guide.
MOTOR	Load is disconnected but motor doesn't rotate	Makes a groaning sound	Faulty switch contact	Adjust the contact.
			Blown fuse	Replace.
			One phase wire of the power supply open	Replace.
			Stator coil open	Repair by rewinding or replacing stator assembly.
			Stator and rotor touching due to bearing housing wear	Replace the bearing and bracket.
		Starts in either direction when turned by hand	Three phase is operating as single-phase	Check the power source with a voltmeter.
		Doesn't make any noise	Stator coil open	Repair by rewinding or replacing stator assembly.
			Outside the motor	Power failure Open connection wire Faulty switch contact Faulty starter contact
				Contact the power company. Check the source wiring.
				Adjust the contact.
	Rotates with the load disconnected, but:	Rotates in the wrong direction	Connection error	Change any two of the three-phase source.
		Fuse blows	Shorted lead wire	Replace.
		Speed doesn't increase	Faulty starter contact	Adjust.

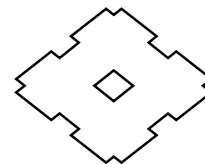


Table 20 Quick Troubleshooting Guide (cont.)

Problem		Possible Cause		Corrective Action
MOTOR (cont.)	Rotates with the load disconnected, but: (cont.)	Groans	Overcurrent Overheating	Repair by winding or replacing stator assembly.
			Overcurrent	Replace the stator winding.
		Makes a high pitched metallic noise	Faulty bearing	Replace the bearing.
	Rotates when the load is disconnected but when the load is connected:	Switch overheats	Insufficient switch capacity	Replace with one having the rated capacity.
			Overload	Drop to the rated load.
		Fuse blows	Insufficient fuse capacity	Replace with one having the rated capacity.
		Overheats	Overload	Drop to the rated load.
			Voltage drop	Consult with the power company.
		Speed suddenly drops	Voltage drop	Consult with the power company.
			Overload	Drop to the rated load.
		Stops	Bearing damaged by overheat	Replace the bearing.
BRAKE	Brake fails to operate		Improper adjustment after reassembly	Adjust again.
	Brake slips (Braking time is too long)	Not wired for fast action		Wire for fast action.
		Foreign matter entrapped in brake lining. Oil on lining surface		Remove foreign matter and take preventive action. Wipe lining surface with a dry cloth.
		Worn brake lining		Adjust brake gap or replace lining.
		Uneven brake gap		Adjust evenly.
		Excessive load		Decrease load or use larger brake.
	Rotor fails to turn	Faulty electric circuit		Check circuit.
		Blow fuse		Replace fuse.
		Only single phase available from three phase power supply		Measure power supply voltage and check for defective circuit.
		Protective device has tripped		Eliminate cause and reset.
		Damaged or burned motor winding		Repair or replace.
		Rust on brake friction surface		Clean brake (lining).
		Gap needs adjustment		Readjust gap.
		Burned bearing		Replace.
		Overload		Check and troubleshoot load and safety device.
	Abnormal noise	Foreign material inside the brakemotor.		Examine inside brakemotor and remove foreign material.
		Damaged bearing		Replace.
		Worn brake lining		Adjust brake gap or replace lining.
		Hub leaf spring is off or damaged		Replace.
		Burned solenoid coil		Replace.
		Damaged rectifier		Replace.
	Trouble under loaded condition	Voltage drop		Raise voltage to rated level.
		Overload		Reduce the load or oversize the brakemotor.
		Improper protective device setting		Adjust protective device.

CONSTRUCTION

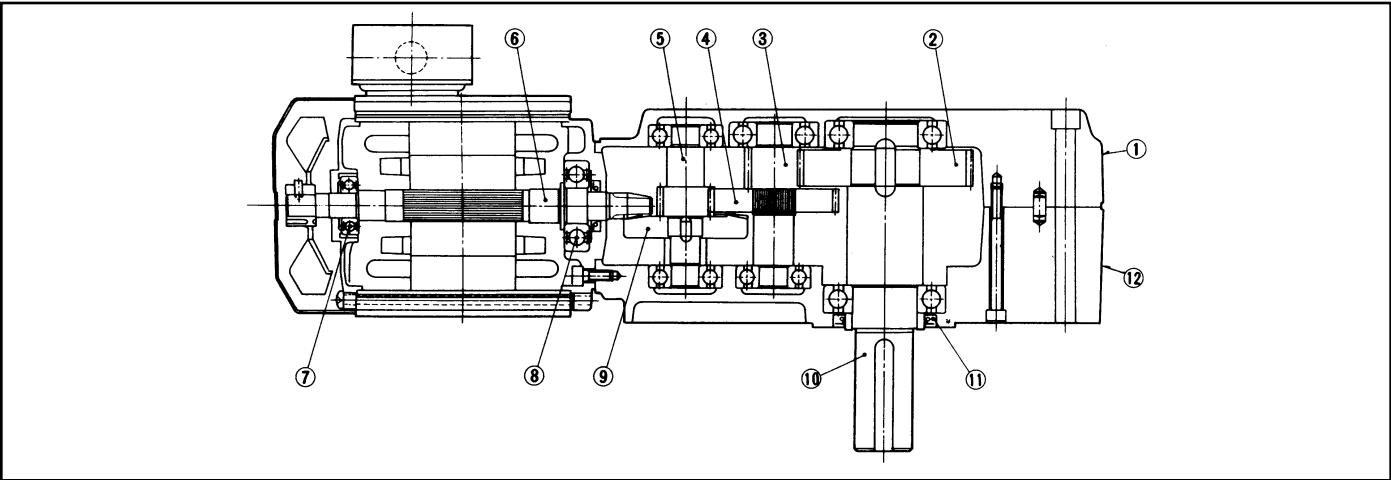


Fig. 21 RNFM series

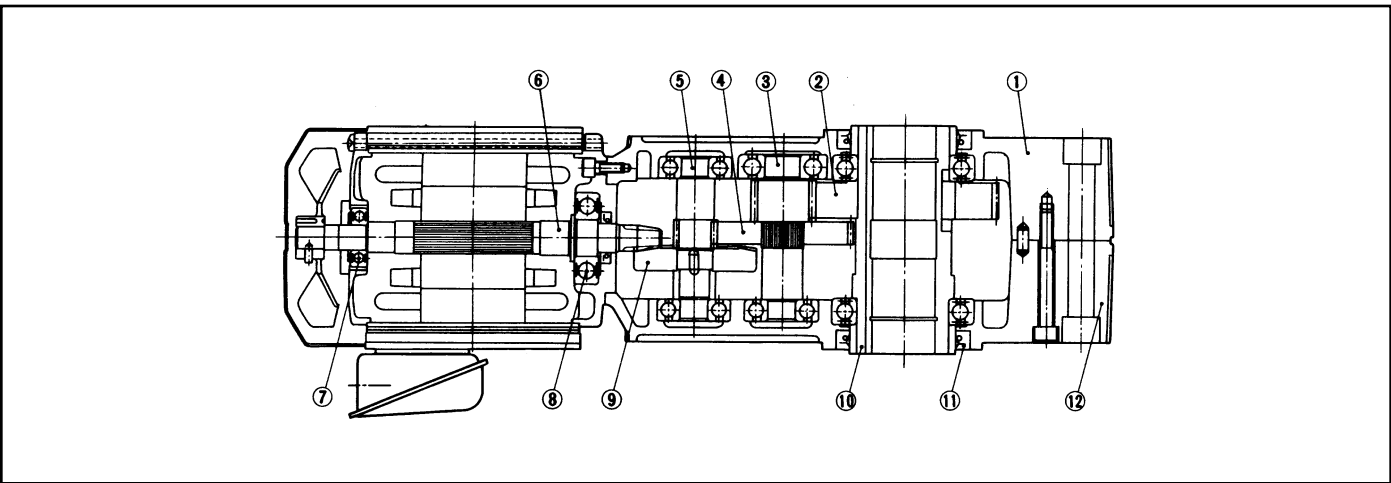


Fig. 22 RNYM series

Table 21 Main Parts

Part No.	Description
1	Case (1)
2	Gear
3	Pinion shaft
4	Gear
5	Pinion shaft
6	Hypoid pinion shaft
7	Bearing metal
8	Bearing metal
9	Hypoid gear
10	Output shaft
11	Oil seal
12	Case (2)



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SM-BEIER

ELECTRICAL VARIABLE SPEED



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AC Drive

NTAC-2000
AC Drive

HELICAL GEAR REDUCER



PARAMAX
Parallel Offset
& Right Angle

SHAFT MOUNT SPEED REDUCER



SM-SHAFT MOUNT
Parallel Offset

DOUBLE ENVELOPING WORM GEAR



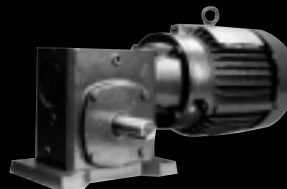
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“SMJ” SERIES BLOWERS



1.0 Applications

Eclipse “SMJ” Blowers are centrifugal blowers that provide low pressure air for industrial combustion systems. They are also used for cooling, conveying, drying, liquid agita-

tion, smoke abatement, vacuum cleaning, fume and dust exhausting, and other applications where air temperatures are under 220°F.

2.0 Installation

2.1 Blower Installation

Install the blower in an accessible location on a level concrete floor or other substantial mounting platform. The top of the blower base should be level to avoid unnecessary wear on the motor bearings. The blower may be bolted to the platform through the mounting holes provided in the blower base. To prevent vibration which can shorten motor bearing and rotor life, install compressed rubber mounting pads between the blower base and floor.

2.2 Blower Environment

If the blower is mounted in a location where excessive airborne solids are present, use an air filter on the blower inlet or connect the inlet to clean outside

air with a pipe no smaller than the blower inlet. Eclipse strongly recommends the use of a combustion air filter suitable for the operating conditions. Blowers can be purchased with filter installed, or filters can be added in the field.

An optional inlet, with an ANSI flange, is available for applications requiring silencers or direct piping.

Provide access to the blowers for inspection and maintenance.

2.3 Filter Mounting

To mount a filter in the field, remove the grill from the fan cover plate. Align the six holes on the back of the filter housing with the corresponding holes on the fan cover plate, as shown in Figure 1 on page 3. Secure the filter housing to the fan cover plate with six self-tapping screws.



Eclipse Combustion

3.0 Piping

3.1 Housing Rotation

SMJ Blowers are available in four different outlet positions. The outlet position that requires the least number of piping bends should be used.

If desired, the outlet position can be changed in the field by disassembling the side plate, rotor, and housing, from base. The housing can then be rotated to the desired outlet position and the blower re-assembled. See Figure 2.

3.2 Minimize Pressure Losses

Air piping should be of ample size, piping runs kept short as possible, and number of bends kept to a minimum in order to avoid excessive pressure losses. As a general rule, when using a blower in a combustion system, keep the air manifold the same size as the blower outlet up to the point of branching off to individual burners. Make sure that all piping is air tight to prevent leakage which causes excessive power consumption.

3.3 Piping Support

Do not use the blower to support the piping. Use independent pipe hangers for this purpose.

3.4 Using Flexible Piping

Use a flexible collar or pipe to connect the blower to the manifold if:

- a) The manifold is 30 feet or longer;
- b) The manifold is vertical and longer than 15 feet;
- c) One blower supplies multiple burners.

3.5 Outlet Flange Specifications

SMJ Blowers are provided with either threaded or flanged outlet, depending on blower size. Flanged outlets are drilled to standard 125# ANSI specifications. Companion flanges are available from Eclipse as optional equipment.

4.0 Electrical Supply

4.1 Check Motor Nameplate

Check the motor nameplate to be sure the motor is correct for the electrical service. Follow the wiring instructions on the motor nameplate. All wiring and fuses must be ample to supply proper voltage and current to the motor. All AC motors should be

equipped with a contactor or magnetic starter having thermal overload protection. Check to see that proper heaters are installed in the motor starter.

4.2 Check The Rotor

On initial startup, check to see that the rotor is turning in the correct direction. **If rotation is wrong, reverse wiring.**

5.0 Maintenance

CAUTION

A blower should never be run with both air inlet and outlet wide open. Proper restriction must be located in either inlet or discharge line to avoid overloading and damaging the motor.

5.2 Clean any accumulation of dirt or dust from the rotor and interior of the blower housing as required. After cleaning, make sure that any remaining dirt does not unbalance the rotor.

5.3 If the rotor fails, carefully check the motor bearings for excessive wear. Worn bearings are usually the cause of rotor failure.

5.1 Protect the motor from excessive heat and overload conditions. See the motor nameplate for temperature specifications.

FIGURE 1—FILTER MOUNTING

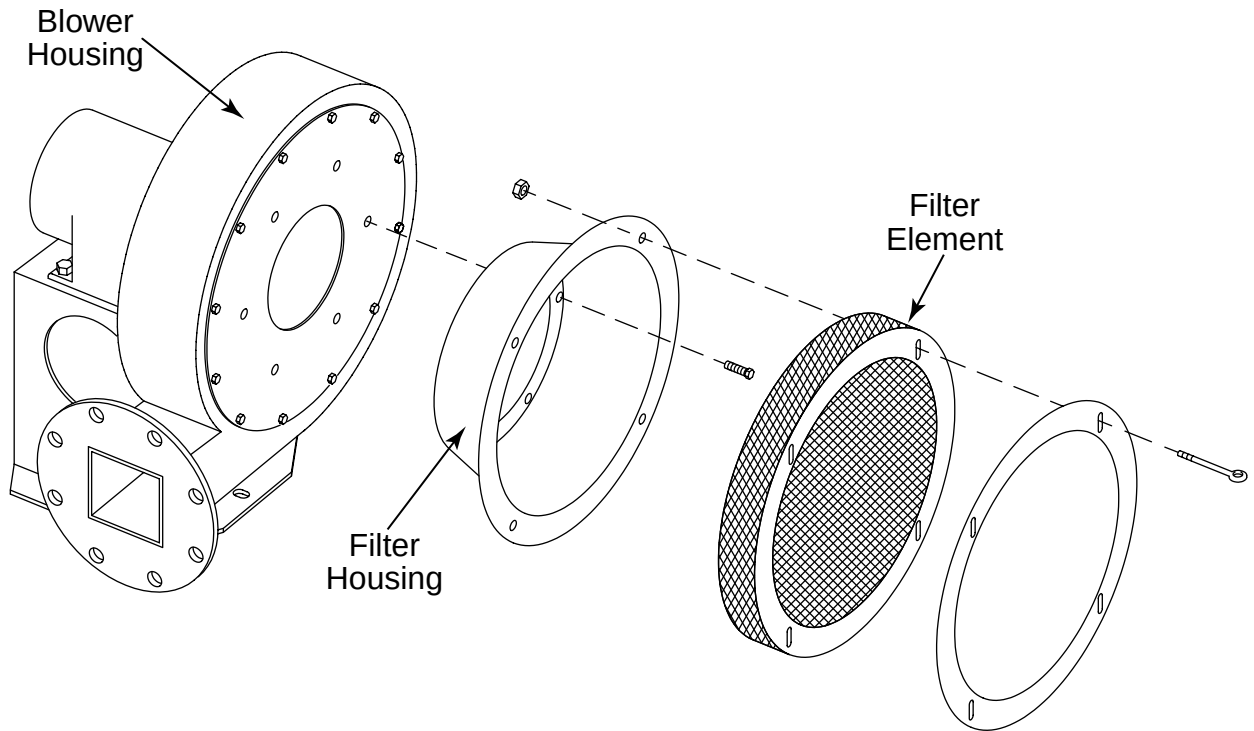
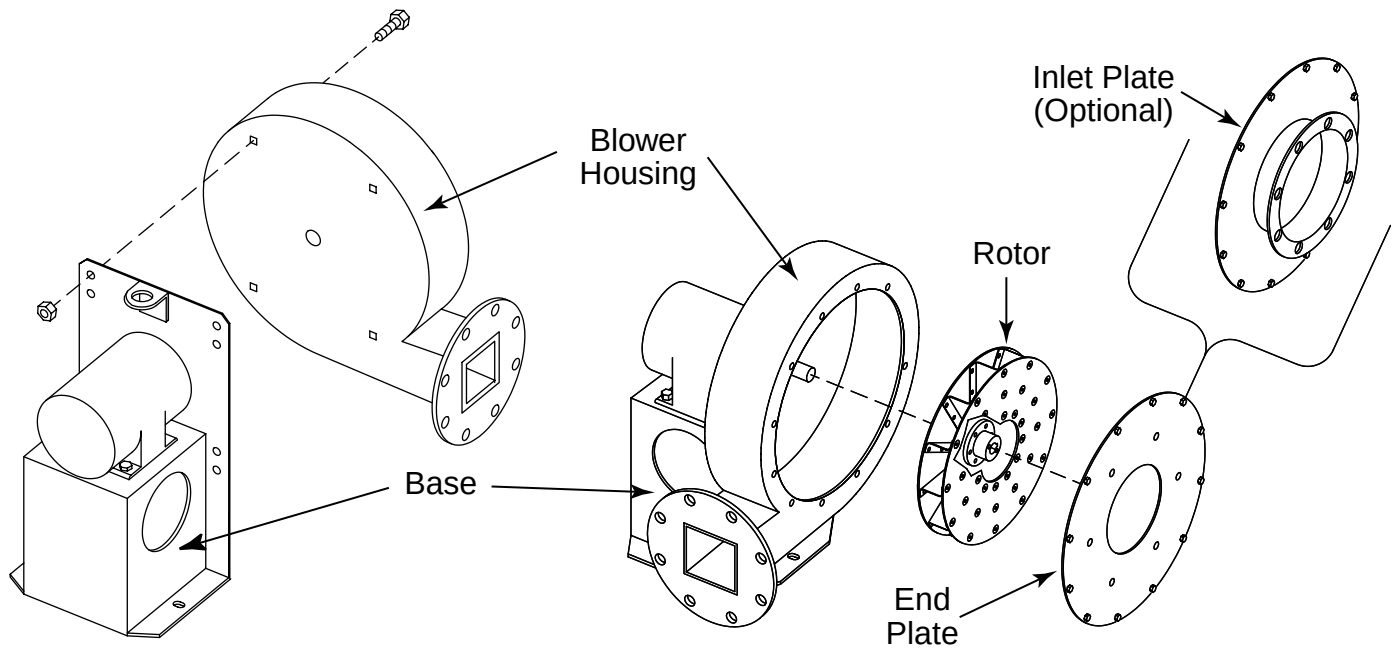


Figure 2—Housing Rotation



ASHWORTH BROS., INC.

WINCHESTER, VIRGINIA
SALINAS, CALIFORNIA

Installation and Operating Recommendations

Balanced Weave Belts

The Manufacturer guarantees quality of workmanship and the metal used in the fabrication of the belt to be as specified. Belts are tested for true travel and rigidly inspected before shipment.

The care used in the installation of the new belt and the observance of good operating practices are insurance of long life and low maintenance cost.

The following suggestions are offered for your guidance:

GENERAL

Friction driven Woven Wire Belts are designed for operation over *straight faced* terminal pulleys. Flanged pulleys will not control side travel and are not recommended, nor should crowned pulleys be used.

Belt tension beyond that required to move the loaded belt should be avoided. Where conditions permit, the drive pulley should be lagged. Snub rolls, pressure or pinch rolls effectively reduce belt tension and are in general use on Wire Belt installations.

Terminal pulleys should be adjusted to be square to the conveyor centerline. To correct belt side travel recheck terminal alignment and make corrections to the belt supports as necessary. Permanent damage to the belt can occur if terminals are "cocked". See Bulletin IC75 for further information on belt tracking.

Supporting rolls must be level and at right angles to the conveyor center line. Belt supporting surface should be checked for freedom from obstructions such as warped hearth plates or protruding framework which might cause belt impingement. Check the belt take-up mechanism to make certain it is functioning properly.

Before installing the new belt make sure none of the spirals have been turned during shipment and that all spirals are flat and free to turn on their proper axis.

TO MAKE BALANCED WEAVE BELTS ENDLESS (Refer to Sketch)

For ease of handling, most belts, because of their length and weight, are shipped in two or more sections. Normal care should be used in uncrating to prevent damage to the fabric.

The same method of joining the sections applies to making the belt endless. Our practice is standard whether the belt is shipped in one continuous length or is made up of two or more sections. *All* belt lengths terminate with a loose spiral. The loose spiral is left hand (counter clockwise winding). The fixed spiral, that is welded to the crimp rod at the opposite end, is right hand (clockwise winding).

Extra crimped connectors and spirals are included with every belt shipped. The spirals have a long end loop to facilitate fixation and the crimped connecting wires are longer than required for the belt width.

TO MAKE ENDLESS—bring the two sections together with the edges of the belt in line. Mesh the end spirals together to permit insertion of the crimped connector wire. Make certain the spirals so joined are of opposite weave—one right, the other left hand.

After inserting the connector wire, close down or knuckle the open ends of the spiral. Cut the crimped connector to the proper length and weld or turn in the protruding ends of the connector.

SURFACE COMBUSTION

The efficient use of energy to minimise environmental impact has encouraged wider use of natural gas, the cleanest fossil fuel. The need for infrared heating and low pollutant emission has caused a renewed interest in surface combustion.

Surface combustion is a gasburning technique in which premixed gas and air burns on a surface layer of a permeable and/or perforated medium. Depending on the intensity, the combustion can occur in two modes:

- Radiant: with the combustion inside the material. The permeable medium then heats to incandescence and releases a portion of the energy input as thermal radiation. Flame colour is red/orange.
- Blue flame: blue flames hover above the surface and release the major part of the energy in a convective way.

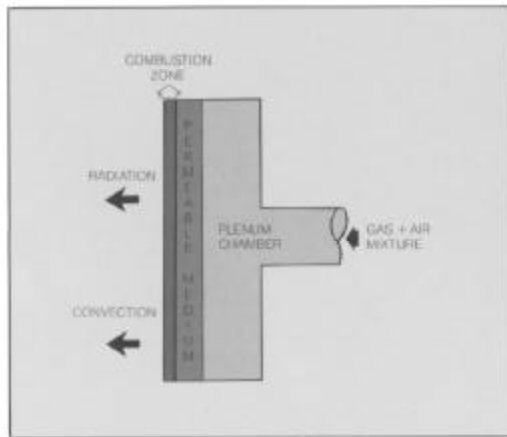


Fig. 1. Surface combustion burner configuration in radiant mode.

HIGH QUALITY STANDARDS FOR PRODUCTION

The Metal Fibre Burner (MFB) mats are produced in accordance with the requirements of the European Quality Standard EN 29001. This Quality Management System secures a continuous and high quality level for all MFB products.

THE METAL FIBRE BURNER

The permeable medium of the MFB consists of fine metal fibres. These can be sintered together to produce a rigid, but highly porous panel or can be converted into a flexible knitted material by means of textile processing.

The porous structure of the sintered product created by randomly laid fibres is shown in Fig. 2. Acotech provides this product with a hole pattern. This perforation leads to a more homogeneous combustion for both radiant and high-intensity blue flame mode and allows for a smooth transition between both modes. The knitted MFB is made from a yarn made from metal fibres. This creative application of textile technology leads to a very flexible burner mat (Fig. 3). The sintered mat will be used for those applications with emphasis on large surface radiant application whereas knitted products will be used for smaller burners of different shapes (cylinders, spheres, cones, ...)



Fig. 2. Sintered MFB (AC product series)

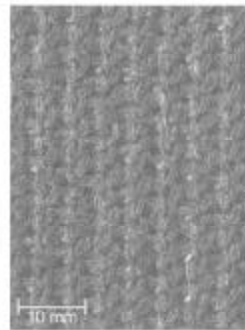


Fig. 3. Knitted MFB (NIT product series)

The fibres for all MFB, produced according to a highly sophisticated process, are made out of Fecralloy, an alloy registered by the U.K. Atomic Energy Authority. This refractory steel was selected for its outstanding oxidation resistance at temperatures over 1000 °C (1830 °F). Of special importance in the alloy composition is the Yttrium element which anchors the protective surface alumina layer to the base metal in a very tenacious way.

ELEMENTS	FECRALLOY
Chromium (Cr)	20.00 %
Aluminium (Al)	5.00 %
Yttrium (Y)	> 0.10 %
Silicon (Si)	0.30 %
Manganese (Mn)	0.08 %
Copper (Cu)	0.03 %
Carbon (C)	0.03 %
Iron (Fe)	Balance

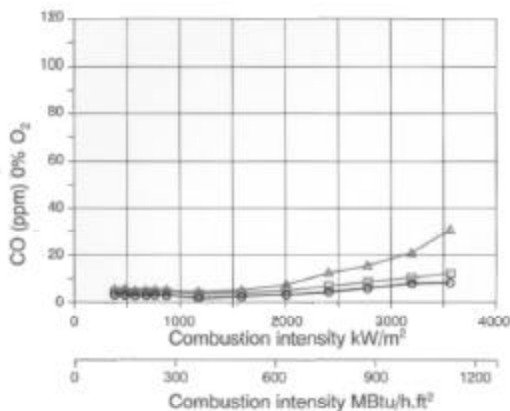
Chemical analysis of Fecralloy.

LOW POLLUTANT EMISSIONS

RADIANT MODE

Radiant mode combustion of the MFB leads to extremely low emissions of NO_x , CO and unburnt fuel components.

As a result of the intimate contact between the gases and the fibres, flame temperature is significantly reduced leading to far lower NO_x levels than other burner technologies.

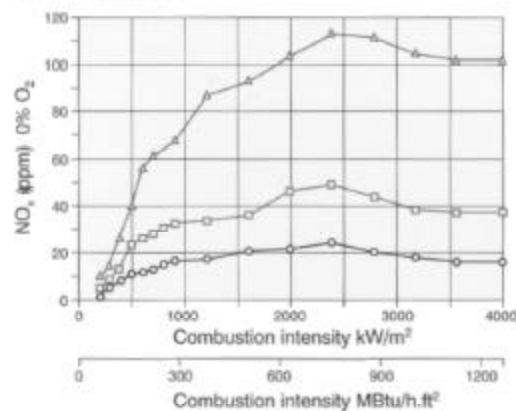


▲ 10% Excess air □ 20% Excess air ○ 30% Excess air

Fig. 4. CO emissions with different excess air ratios and firing rates.

BLUE FLAME MODE

Blue flame mode combustion can still occur at very low NO_x , CO emissions when using MFB. In this mode higher excess air ratios are suggested. This mode occurs at intensities higher than 1000 kW/m^2 (300 MBtu/h.ft^2) and can support intensities over 5000 kW/m^2 (1600 MBtu/h.ft^2). Fig. 4 and Fig. 5 show the evolution of the NO_x , CO emissions as a function of the combustion intensity of the burner and excess air ratios.



▲ 10% Excess air □ 20% Excess air ○ 30% Excess air

Fig. 5. NO_x emissions with different excess air ratios and firing rates.



Multitude of Metal Fibre Burners for different applications developed by Acotech.

ACOTECH

COMBUSTION CHARACTERISTICS

RADIATION EFFICIENCIES

Radiation from surface combustion burners originates from two sources: emissions from the heated surface and emissions from the hot combustion gases leaving the surface. The high radiant flux from the MFB can be expressed as a portion of the total energy output of the burner (Fig. 6). These radiation levels can be significantly enhanced by preheating the combustion air for example by utilising the waste flue gas heat. In this way radiative efficiencies over 50% may be readily achieved. Similar efficiency increases can be obtained by putting a metal wire grid in front of the burner surface.

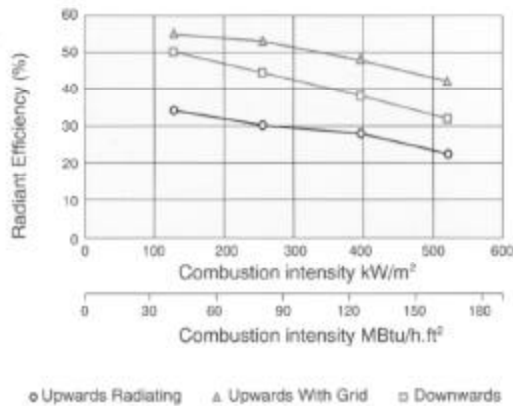
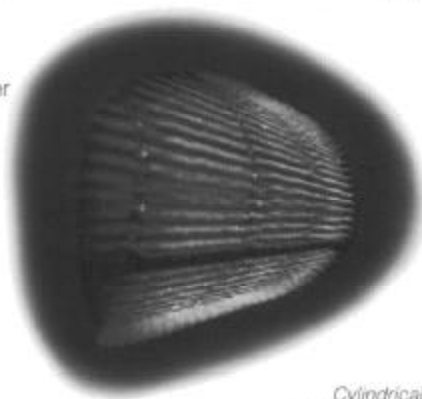


Fig. 6. Radiation efficiency versus combustion intensity (MFB emissivity: 0.7) (Courtesy: Gaz de France)

The surface temperature increases with the firing rate from 750 °C (1380 °F) at 100 kW/m² (32 MBtu/h.ft²) up to 1000 °C (1830 °F) at 500 kW/m² (160 MBtu/h.ft²). When the burner is operated in a closed environment, surface temperatures will be higher and radiative power will increase. The maximum longtime operating temperature for the MFB is 1100 °C (2010 °F).



Cylindrical Low NO_x Burner
(ø 800 mm/31" - L.3000/118") for
a 10 MW (32 MM Btu/h)
oil field steam generator
(By courtesy of Alzeta, USA)

FIRING RATE/HEATING EFFICIENCY

The minimum firing rate of the MFB is 100 kW/m² (32 MBtu/h.ft²). Below 100 kW/m² (32 MBtu/h.ft²), the combustion reaction is unstable and may extinguish. Up to 800 kW/m² (255 MBtu/h.ft²), the burner operates in the radiant mode. Between 800 and 900 kW/m² (255 and 285 MBtu/h.ft²), the flame base starts to raise above the surface. Above 900 kW/m² (285 MBtu/h.ft²), the burner is no longer radiant and operates in the blue flame mode. In this blue flame mode, some burner design types have been operated up to 5000 kW/m² (1600 MBtu/h.ft²). This results into high turndown capabilities.

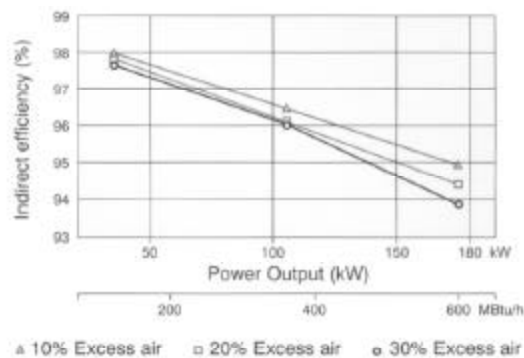


Fig. 7. Indirect efficiency of an MFB 200 kW (680 MBtu/h) cylindrical burner in a condensing boiler (as reported by Gaswärme-Institut Essen). For the design used, 200 kW corresponded to a surface burning intensity of 4000 kW/m² (1260 MBtu/h.ft²).

At these high intensities a 200 kW (680 MBtu/h) cylindrical burner is not longer than 18 cm (7") with a diameter of 10 cm (4").

The possibility to achieve high burner capacities in such a compact way opens totally new design perspectives. In Fig. 7 figures show that combined with condensing boilers MFB burners can achieve very high combustion efficiencies.

ENDURANCE AND PERFORMANCE EVOLUTION OVER TIME

To verify specifications and durability of the material, a large multiburner test facility has been constructed at the Grand Couronne Research Centre of Shell Recherche S.A. Fig. 8 shows the facility which can test up to 24 burner mats of 15 cm square size simultaneously. The system runs automatically 24 hours per day with an 8 minutes on and 2 minutes off cycle to accelerate cyclic thermal ageing of the material. Heat reflectors in front of the burner are also used to shorten the necessary life test period. The combination of severe heat exchanger

conditions - up to 1000 °C (1832 °F) heat reflector temperature - allied to the extreme thermal cycling rate enable burner wear to be monitored over a period of months rather than many years. After 36 months of operation the burner mats show no significant deterioration. The thickness of the material was unchanged, with no traces of wear or cracks. These studies confirmed the exceptional durability of metal fibre burners for a wide range of applications. The endurance testing of newer products is still going on.

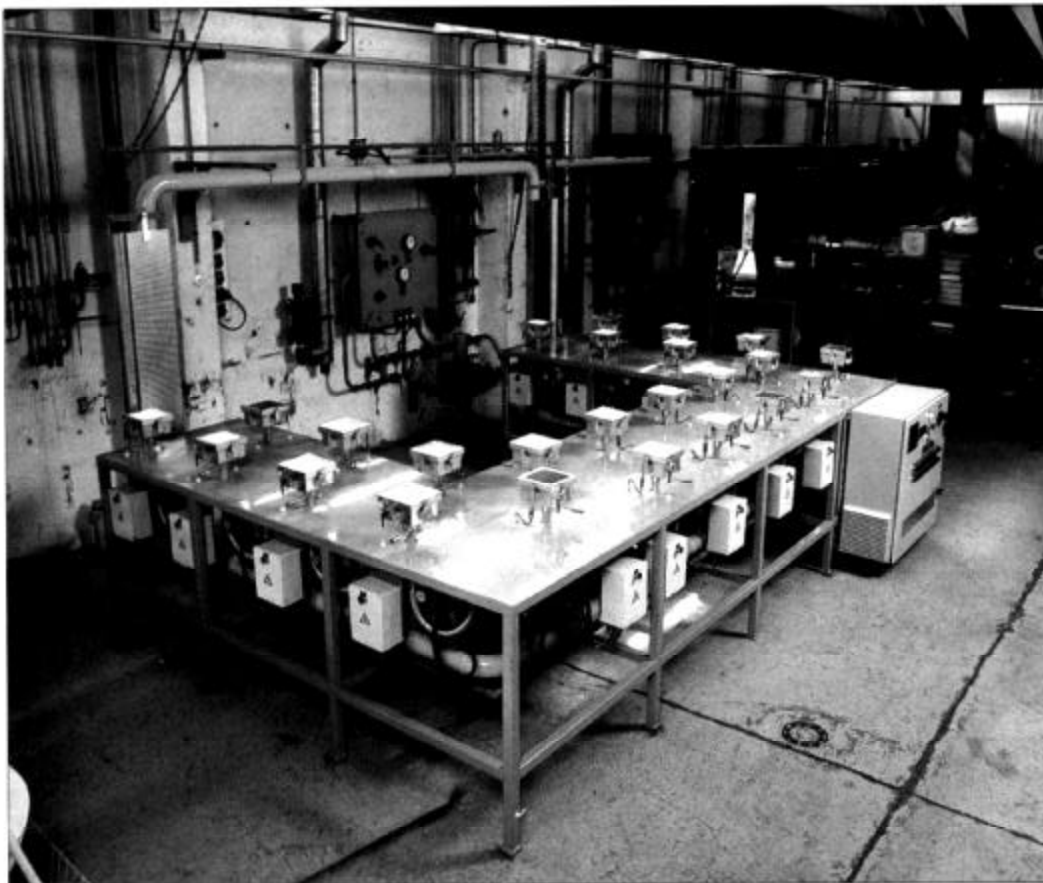


Fig. 8. Metal Fibre Burner endurance test bench at the Grand-Couronne Research Centre of Shell Recherche S.A. (Rouen, France).

MINS 1052-5



*Metal Fibre Burner for the annealing of crystal glass
(By Courtesy of Tokyo Gas, Japan)*

APPLICATIONS

MFB has proven its worth in a whole range of commercially available products. Because of its unique characteristics, MFB has allowed burner manufacturers incorporating this technology in their products to gain significant market shares. MFB has been particularly successful in the following markets:

PAPER DRYING

Because of MFB's extremely low thermal inertia and ruggedness, the material is second to none when it comes to effectively and safely applying gas for IR drying. A growing number of paper mills are specifying MFB for their coating curing dryers.

OTHER DRYING APPLICATION

MFB is globally being applied for the drying of textiles, ceramic tiles, inks, paints and coatings.

RESIDENTIAL AND INDUSTRIAL LOW NO_x BOILERS AND WATERHEATERS

More governments are specifying maximum levels of NO_x and CO emissions from residential heating appliances. To respond to these requirements, the advanced boiler and burner manufacturers have been switching their top end equipment over to MFB (over 100,000 units sold per year). MFB continuously gains over other low NO_x technologies because the material has proven its endurance and can be easily transformed. MFB burners can be flexibly modulated over a wide range (high turndown ratio) and operate safely.

GLASS INDUSTRY

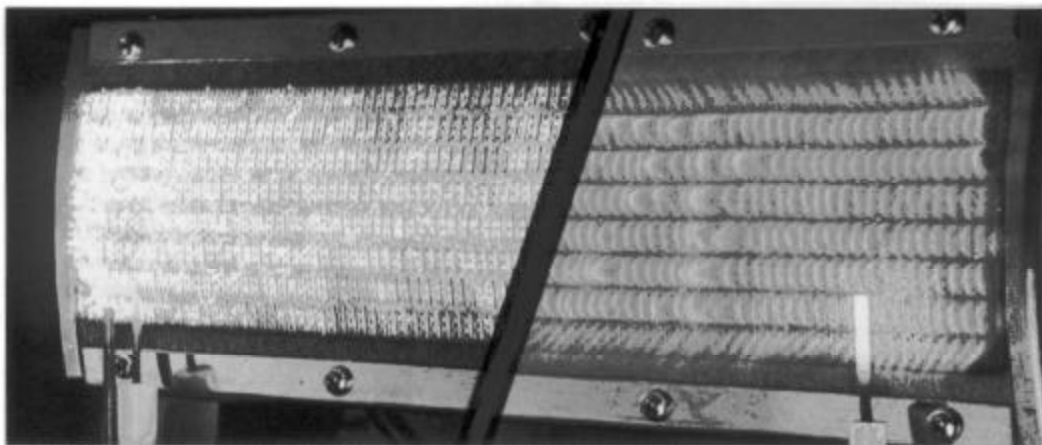
MFB is used at present in furnaces for automotive glass bending because of the possibility to modulate quickly the radiant heat flux in function of the shape of the window. Other applications in the glass industry are the annealing of crystal glass, medical tubes, street lamps and cathode ray tubes.

FOOD PROCESSING

In the food industry MFB is being used in fryers, pie grillers, and biscuit ovens. The material presents no health hazards and is extremely reliable.

NEW DEVELOPMENTS

Applications are being developed in a continuous way. We see new entries in the steel industry, aluminium casting and processing industry.



MFB mat on ported deck gas burner. Modulating between the radiant and the blue flame mode. Large series of gas boiler units are equipped with this technology. (By courtesy of Furigas B.V., The Netherlands).

OPERATIONAL SAFETY

PROTECTION FROM FLASHBACK

Even in tough confined operating conditions with the MFB surface heated up to 1150 °C (2100 °F), insulation properties of the MFB assure that no self-ignition will occur, even of such gases with low igniting temperature such as propane, butane or hydrogen containing gases. In all the years MFB's have been in service no flashback incident has occurred.

RESISTANCE TO THERMAL SHOCKS

The MFB being 100% metal, resists thermal shocks even of an extreme nature. Thus water splashed on the MFB surface whilst firing evaporates without leaving any trace of damage to the burner mat.

RESISTANCE TO MECHANICAL SHOCKS

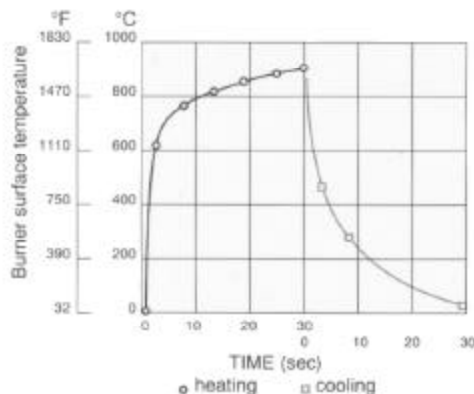
The MFB is extremely robust. No special precautions are needed for handling or mounting. It is common practice in demonstration tests to hit the burner surface with a hammer without any effect on mechanical integrity or burner operation.

FAST RESPONSE

Because of the highly porous nature of the mat as well as the thin combustion zone, the MFB is operating in radiant mode a few seconds after ignition. This is a major benefit for modulation and heating accuracy.

FAST COOLDOWN

This is a crucial point in fire prevention. If the burner is shut down with the airfan switched on, the cooling down of the MFB is so fast that a few seconds after shutdown the burner surface can be touched with bare hands.

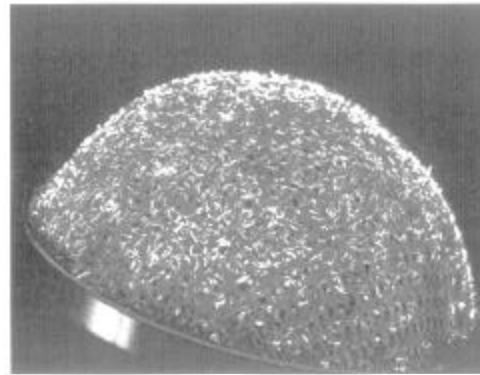


Heating/cooling dynamic of MFB in open air.

RELIABLE AND EFFICIENT

UNIFORM COMBUSTION

At any firing rate the burner shows homogeneous combustion and uniform radiant and convective heat output.



Spherical Metal Fibre Burner equipped with knitted MFB mat.

CORROSION RESISTANCE

AT HIGH TEMPERATURE

The sophisticated FeCrAlloy steel used to manufacture the MFB is extremely resistant to oxidation and corrosion. This provides the MFB with exceptional durability.

LOW PRESSURE DROP

Depending on the firing rate and the eventual perforation of the material, the pressure drop of the MFB varies between 1 and 40 mm water column.

EASY MAINTENANCE

When by accident organic materials are spilled on the surface of the MFB, they can easily be burnt off by pyrolysis when firing the burner. Inorganic products can be removed by cleaning the burner surface with an airjet or waterjet. The mechanical coherence of the fibre material is strong, so even rough cleaning treatments are possible without disturbing the MFB mechanical integrity.

VERSATILITY

The MFB material may be treated as a metal material. Machining, drilling, bending, welding operations are possible. As a consequence, the MFB mat can be used to create the most versatile burner shapes such as cylinders, cones, diamonds, boxes, spheres.